

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)	(i)	Acceptable line of best fit by eye – bisects data points at (0.4,0.925) and (0.5,1.25) (1) Doesn't need to pass through zero but needs to be drawn back to an axis Gradient calculated. Accept range: 2.3 – 2.5 (1) No unit penalty		2		2	2	2
		(ii)	Unit of force (Newton) seen as kg m s^{-2} (1) $\frac{\text{m s}^{-2}}{\text{kg m s}^{-2}}$ seen [= kg^{-1}] (1) Accept: Gradient = $\frac{a}{F} = \left(\frac{\text{N}}{\text{kg N}} \right)$ argument award 2 marks	1	1		2	1	2
		(iii)	$\frac{1}{\text{gradient}} = 0.40 - 0.44$ [kg] (1) (ecf on gradient). Accept correct use of data points. $0.42 = M + 0.06$ (1) $M = 0.36$ [kg] (1) Accept range: 0.34 – 0.38 [kg]			3	3	3	3
		(iv)	Quality: Points lie close to line of best fit so quality acceptable or reference to possible anomaly at 0.5 N (1). Don't accept reference to measuring apparatus. Sufficiency: Too few points plotted or more needed between 0.0 and 0.4 N or results missing at e.g. 0.1 N or 0.3 N (1)			2	2		2
	(b)	(i)	Weight of glider = $9.81 \times M$ (= 3.1 – 3.7) N (ecf on M) (1) Force due to air must be same answer as weight of glider (1)		2		2	1	
		(ii)	Forces act on the same body (1) Forces are not of the same kind or details provided e.g. contact/electrostatic and gravitational (1)	2			2		
			Question 3 total	3	5	5	13	7	9